

AR Weight Horserace

Instrument Strength Across Weighting Schemes

April 2026

Summary of Results

- ▶ 80 specifications: 4 weights $\times$ 5 tier specs $\times$ 4 control ladders; all estimated successfully
- ▶ Outcome: log municipal GDP; H_0 : shift-share instruments jointly zero in reduced form
- ▶ Controls ladder: None $\rightarrow$ FE only $\rightarrow$ FE + Aggregate Exposure $\rightarrow$ FE + Sector Exposure

Weighting Scheme	Specs rejecting H_0 at 5% (FE + Sector Exp.)	Best tier(s) at FE + Sector Exposure
Employment Share (Municipal)	4 of 5	Mayor + Governor; Mayor + Gov + President
Binary Affiliation (Municipal)	5 of 5	Governor
Ownership Share (Municipal)	4 of 5	Governor; Mayor + Governor
Ownership Share (Legacy)	3 of 5	Governor; Mayor + Governor

Employment Share (Municipal)

	No Controls	FE Only	FE + Agg. Exposure	FE + Sector Exposure
Mayor	47.41*** [p=0.000]	2.92** [p=0.020]	2.45** [p=0.044]	2.30* [p=0.056]
Governor	69.21*** [p=0.000]	3.11** [p=0.014]	4.15*** [p=0.002]	3.80*** [p=0.004]
Mayor + Governor	36.56*** [p=0.000]	4.47*** [p=0.000]	4.79*** [p=0.000]	4.50*** [p=0.000]
Mayor + President	47.18*** [p=0.000]	2.22** [p=0.023]	2.07** [p=0.035]	1.97** [p=0.046]
Mayor + Governor + President	55.31*** [p=0.000]	3.31*** [p=0.000]	3.51*** [p=0.000]	3.32*** [p=0.000]

4 of 5 tier specs reject H_0 at 5% under sector exposure controls.

Binary Affiliation (Municipal)

	No Controls	FE Only	FE + Agg. Exposure	FE + Sector Exposure
Mayor	70.26*** [p=0.000]	3.46*** [p=0.008]	3.36*** [p=0.009]	3.35*** [p=0.009]
Governor	88.30*** [p=0.000]	5.05*** [p=0.000]	6.22*** [p=0.000]	5.79*** [p=0.000]
Mayor + Governor	53.94*** [p=0.000]	5.41*** [p=0.000]	5.98*** [p=0.000]	5.78*** [p=0.000]
Mayor + President	45.87*** [p=0.000]	2.40** [p=0.014]	2.41** [p=0.014]	2.42** [p=0.013]
Mayor + Governor + President	42.42*** [p=0.000]	4.45*** [p=0.000]	4.88*** [p=0.000]	4.75*** [p=0.000]

5 of 5 tier specs reject H_0 at 5% under sector exposure controls.

Ownership Share (Municipal)

	No Controls	FE Only	FE + Agg. Exposure	FE + Sector Exposure
Mayor	120.26*** [p=0.000]	6.14*** [p=0.000]	2.19* [p=0.068]	2.27* [p=0.059]
Governor	163.08*** [p=0.000]	4.99*** [p=0.001]	10.67*** [p=0.000]	10.60*** [p=0.000]
Mayor + Governor	76.56*** [p=0.000]	7.01*** [p=0.000]	7.76*** [p=0.000]	7.66*** [p=0.000]
Mayor + President	157.71*** [p=0.000]	4.35*** [p=0.000]	2.50** [p=0.010]	2.38** [p=0.015]
Mayor + Governor + President	106.45*** [p=0.000]	5.54*** [p=0.000]	5.77*** [p=0.000]	5.83*** [p=0.000]

4 of 5 tier specs reject H_0 at 5% under sector exposure controls.

Ownership Share (Sector-Normalized, Legacy)

	No Controls	FE Only	FE + Agg. Exposure	FE + Sector Exposure
Mayor	123.18*** [p=0.000]	3.46*** [p=0.008]	0.56 [p=0.689]	0.65 [p=0.623]
Governor	99.93*** [p=0.000]	4.48*** [p=0.001]	12.82*** [p=0.000]	17.12*** [p=0.000]
Mayor + Governor	73.13*** [p=0.000]	4.62*** [p=0.000]	7.03*** [p=0.000]	8.96*** [p=0.000]
Mayor + President	66.77*** [p=0.000]	2.91*** [p=0.003]	1.85* [p=0.063]	1.09 [p=0.365]
Mayor + Governor + President	52.22*** [p=0.000]	3.85*** [p=0.000]	5.85*** [p=0.000]	6.52*** [p=0.000]

3 of 5 tier specs reject H_0 at 5% under sector exposure controls.